

1 Positive Scalar Curvature

1.1 Overview

brief definition, maybe intuition and obstructions/background etc. based on overview paper

psc is somehow sweet spot for rigidity and flexibility

Probably the most important existence Theorem for positive scalar curvature metrics... See Appendix for a brief review of surgery theory.

Theorem 1.1 (Schoen, Yau 1979, Gromov, Lawson, 1980). (*Surgery Theorem*)

In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has been subject of research especially in the context of the h -cobordism theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

Theorem 1.2 (Wall, 1971). (*Wall Theorem*)

Combining Gromov, Lawson and Wall we see that: M_0 is psc, $M_1 \rightarrow W$ 2-connected then M_1 psc. In fact one can use the Theorem of Wall to obtain an even stronger result:

Theorem 1.3 (extending psc to cobordism, Georg, Gaja). ...

So in view of the above discussion, one strategy to construct a psc metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some psc manifold M_0 to M_1 and then modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is. In order to formulate these we recall the following definition.

Definition 1.4 (Spin and Totally non spin). ...

We now have the following two existence theorems for psc metrics. One for the spin case and one for the totally non-spin case.

Theorem 1.5. *Let W^{n+1} be a cobordism from M_0 to M_1 , $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

Theorem 1.6. *Let W^{n+1} be an oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

Proof of Theorem 1.6 and 1.5

In the main construction in section ... we will invoke Theorem 1.6 to construct a psc metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 1.7. (i) *The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

(ii) *A manifold containing a totally non-spin submanifold of codimension 0 is totally non-spin.*

(iii) *If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \text{int}(M)$ in the interior of M .*

Remark 1.8. All statements of the above lemma remain valid if “totally non-spin” is replaced by “non-spin”. The corresponding proofs are just simpler versions of the arguments given below.

Proof. For (i) let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_1, \quad \pi_2: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_2$$

the projections and let $\iota: \tilde{M}_1 \rightarrow \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M . Let $\iota: N \rightarrow M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota}: T\tilde{N} \rightarrow T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{array}{ccc} T\tilde{N} & \xrightarrow{d\tilde{\iota}} & T\tilde{M} \\ \downarrow & & \downarrow \\ \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \end{array}$$

Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) assume without loss of generality that M is connected. Denote by $B^n := \text{int}(D^n)$ the open ball. Then $M - D^n$ is the interior of $M - B^n$ from which we conclude that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. This follows using an analogous argument as in (ii) by noting that for a manifold X with boundary the universal cover of $\text{int}(X)$ is given by $\text{int}(\tilde{X})$ and the inclusion $\text{int}(\tilde{X}) \rightarrow \tilde{X}$ is a homotopy equivalence. We will now show that $M - B^n$ is totally non-spin. Let $p: \tilde{M} \rightarrow M$ be the universal cover of M and write $M_0 := M - B^n$. We claim that the universal cover of M_0 agrees with the covering

$$\hat{M}_0 = \tilde{M}|_{M_0} = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of p . From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since B^n is simply connected. If the fiber F is finite we can apply the Seifert-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk B^n removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi: S^1 \rightarrow \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi: D^2 \rightarrow \tilde{M}$ restricting to α on the boundary. We now show that Φ is homotopic rel S^1 to some map $\Phi_1: D^2 \rightarrow \tilde{M}$, such that the image of Φ_1 is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq M$ be the center of the embedded disk. After a preliminary homotopy we may assume that Φ is transversal to $p^{-1}(0)$. The crucial observation is that, since $n \geq 3$, this is equivalent to requiring that Φ does not intersect $p^{-1}(0)$. Now note that S^{n-1} is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r: \tilde{M} - p^{-1}(0) \rightarrow \hat{M}_0$$

be such a retraction and let

$$r_t: \tilde{M} - p^{-1}(0) \rightarrow \tilde{M} - p^{-1}(0)$$

be a homotopy rel S^1 from the identity to r (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy rel S^1 from Φ to some map Φ_1 whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of M_0 is given by

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F).$$

Finally, the Mayer-Vietoris sequence of the pushout

$$\begin{array}{ccc} S^{n-1} \times F & \longrightarrow & D^n \times F \\ \downarrow & & \downarrow \\ \tilde{M} - (B^n \times F) & \xrightarrow{\iota} & \tilde{M} \end{array}$$

where all maps are inclusions, then shows that ι is injective on $H^2(-; \mathbb{Z}_2)$. Therefore

$$w_2(T\hat{M}_0) = w_2(\iota^*T\tilde{M}) = \iota^*w_2(T\tilde{M}) \neq 0.$$

□

A standard example of a non-spin manifold is $\mathbb{C}P^2$. Since $\mathbb{C}P^2$ is simply connected this is also an example of a totally non-spin manifold, which we will appeal to in section ...

Lemma 1.9. *$\mathbb{C}P^2$ is non-spin.*

(proof)

A Appendix

A.1 A Short Introduction to Surgery

Surgery Theory provides a way to change the topology of a manifold in a controlled way. The basic idea is the following simple observation. A product $S^k \times S^{n-k-1}$ of spheres can be considered as the boundary of both $S^k \times D^{n-k}$ and $D^{k+1} \times S^{n-k-1}$. This observation has the following consequence. Let M^n be a smooth n -dimensional manifold and let $F: S^k \times D^{n-k} \rightarrow M$ be an embedding into the interior of M . Then we can remove the interior of the image of F resulting in a manifold

$$\tilde{M} := M - \text{int}(\text{im}(F)).$$

To $\tilde{M}$ we can attach $D^{k+1} \times S^{n-k-1}$ along the boundary component $F(S^k \times S^{n-k-1}) \subseteq \tilde{M}$ to obtain a manifold M' . More precisely we define M' as the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \xrightarrow{F|_{S^k \times S^{n-k-1}}} & \tilde{M} \\ \downarrow \text{incl.} & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & M' \end{array}$$

Up to diffeomorphism there exists a unique smooth structure on M' , such that the topological embeddings $\tilde{M} \rightarrow M'$ and $D^{k+1} \times S^{n-k-1} \rightarrow M'$ are smooth embeddings, see [Milnor1965, Theorem 1.4]. We say that the manifold M' is obtained from M by surgery along F . Intuitively we remove a thickened k -sphere from M and glue in its place a thickened $(n-k-1)$ -sphere. Accordingly we refer to the surgery along F as a surgery of dimension k or simply a k -surgery.

A different, yet closely related concept, is the theory of cobordisms. Two closed n -dimensional manifolds M and M' are called *cobordant* if there exists a compact manifold W^{n+1} whose boundary is the disjoint union of M and M' . The manifold W is called a cobordism from M to M' . Since any closed manifold is cobordant to itself via the trivial cobordism $M \times [0, 1]$ and cobordisms can be glued together along common boundary components, the relation of cobordism defines an equivalence relation on the class of closed n -dimensional manifolds. One way to construct a new cobordism out of a given one is by means of *handle attachments*. Suppose we are given an embedding $\phi: S^k \times D^{n-k} \rightarrow \partial W$. Then we can form a new manifold W' by attaching a so called $(k+1)$ -handle $D^{k+1} \times D^{n-k}$ to W along ϕ , that is we form the pushout

$$\begin{array}{ccc} S^k \times D^{n-k} & \xrightarrow{\phi} & W \\ \downarrow \text{incl.} & & \downarrow \\ D^{k+1} \times D^{n-k} & \longrightarrow & W' \end{array}$$

We also refer to $(k+1)$ as the *index* of the handle. Some care needs to be taken to define a smooth structure on W' , see [Milnor, Morse Theory]. It turns out that any cobordism W from M to M' can be obtained from the trivial cobordism $M \times [0, 1]$ by attaching finitely many handles [REFERENCE]. Now the connection between surgery theory and cobordisms is the following fundamental result, which can be shown by means of Morse theory.

Theorem A.1 (Milnor1965, Theorem 3.13 and Theorem 3.14 + other source!!). *Let M and M' be closed manifolds of the same dimension. Then M' is obtained from M by a surgery of dimension k if and only if there exists a cobordism W from M to M' containing only a single handle of index $k+1$, that is W is obtained from the trivial cobordism $M \times [0, 1]$ by attaching a single $(k+1)$ -handle.*

One common usecase of surgery is to modify a given manifold M in order to simplify its homotopy groups. Essentially given some embedding $f: S^k \rightarrow M$ we would like to modify the topology of M , such that f becomes contractible. We could just glue in a disk D^{k+1} along f , however this would destroy the manifold structure. So instead we “thicken” the embedded sphere by extending f to an embedding $F: S^k \times D^{n-k} \rightarrow M$, remove this thickened sphere and glue in a whole family of disks $D^{k+1} \times S^{n-k-1}$, i.e. we perform a k -surgery. This process is usually referred to as “killing” the homotopy class $[f]$ by surgery. The fundamental obstruction to performing such a surgery is the existence of the embedding F , which exists if and only if the normal bundle of f is trivial. The following Proposition now formalizes this idea of killing homotopy classes by surgery.

Proposition A.2. *Let W^n be a smooth manifold with basepoint $w_0 \in W$, $M \subseteq \partial W$ a boundary component with basepoint $x_0 \in M$ and $2(k+1) \leq n$.*

- (i) *Let $\alpha_1, \dots, \alpha_l \in \pi_k(W, w_0)$ be classes, which are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l -many k -surgeries away from the point w_0 together with an epimorphism*

$$\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$$

containing the $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by $\alpha_1, \dots, \alpha_l$ in its kernel. Moreover

$$\pi_i(W, w_0) \cong \pi_i(W', w_0)$$

for all $i \leq k-1$.

- (ii) *Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_k(W, M, x_0)$ be classes, which lift to classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$ under the map $\pi_k(W, x_0) \rightarrow \pi_k(W, M, x_0)$ induced by the inclusion. Again assume that $\alpha_1, \dots, \alpha_l$ are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l -many k -surgeries together with an epimorphism*

$$\pi_k(W, M, x_0) \twoheadrightarrow \pi_k(W', M, x_0)$$

containing the $\mathbb{Z}[\pi_1(M, x_0)]$ -module generated by $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ in its kernel. Moreover

$$\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$$

for all $i \leq k-1$.

In the proof of the Proposition we will make use of the following basic lemma.

Lemma A.3. *Let X be a topological space and let $H: S^k \times [0, 1] \rightarrow X$ be a homotopy from some map $f: S^k \rightarrow X$ to some map $g: S^k \rightarrow X$. Fix a basepoint $s_0 \in S^k$ and consider the path $\gamma(t) := H(s_0, t)$ from $x_0 := f(s_0)$ to $y_0 := g(s_0)$. Then the change of basepoint isomorphism*

$$b_\gamma: \pi_k(X, x_0) \rightarrow \pi_k(X, y_0)$$

induced by γ maps the class $[f]$ to the class $[g]$.

Proof. We represent the classes $[f]$ and $[g]$ by maps $D^k \rightarrow X$... □

Proof of Proposition A.2. We first prove the absolute version (i) of the Proposition in three steps.

Step 1. We start by explaining how to kill a single homotopy class by surgery. Suppose that $w_0 \in W$ is an interior point. Let $\beta \in \pi_k(W, w_0)$ be a class represented by an embedding $f: (S^k, s_0) \rightarrow (W, w_0)$ for a specified basepoint $s_0 \in S^k$, which extends to an embedding $F: S^k \times D^{n-k} \rightarrow W$ into the interior of W . Let

$$\tilde{W} := W - \text{int}(\text{im}(F))$$

and define W' by the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

as discussed above. Since the inclusion $S^k \times S^{n-k-1} \rightarrow D^{k+1} \times S^{n-k-1}$ is a cofibration it follows that the above square is a homotopy pushout, hence the map $\tilde{W} \rightarrow W'$ has the same connectivity as $S^k \times S^{n-k-1} \rightarrow D^{k+1} \times S^{n-k-1}$, which is k -connected. Similarly, the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ S^k \times D^{n-k} & \longrightarrow & W \end{array}$$

implies that $\tilde{W} \rightarrow W$ is $n - k - 1$ -connected. Choose any point $d_0 \in \partial D^{n-k} = S^{n-k-1}$ and set $w'_0 := F(s_0, d_0)$. We now construct an epimorphism $\Phi_k: \pi_k(W, w_0) \rightarrow \pi_k(W', w'_0)$ mapping the class β to 0. Choose a path $\lambda: [0, 1] \rightarrow D^{n-k}$ from the center point $0 \in D^{n-k}$ to d_0 and let

$$\eta(t) := F(s_0, \lambda(t)).$$

Then η is a path from w_0 to w'_0 and we can consider the change of basepoint isomorphism

$$b_\eta: \pi_k(W, w_0) \rightarrow \pi_k(W, w'_0)$$

induced by η . From the preceding arguments it follows, that the map $\pi_i(\tilde{W}, w'_0) \rightarrow \pi_i(W, w'_0)$ induced by the inclusion is an isomorphism for $i \leq k$, since $2(k+1) \leq n$. Thus we can define maps

$$\Phi_i: \pi_i(W, w_0) \xrightarrow{b_\eta} \pi_i(W, w'_0) \xleftarrow{\cong} \pi_i(\tilde{W}, w'_0) \rightarrow \pi_i(W', w'_0).$$

By the above discussion Φ_i is an isomorphism for $i \leq k - 1$ and an epimorphism for $i = k$. We now show that Φ_k maps the class β to 0. Define

$$\tilde{f}: (S^k, s_0) \rightarrow (W, w'_0), \quad y \mapsto F(y, d_0).$$

Then $\tilde{f}$ is freely homotopic to f via the homotopy $(y, t) \mapsto F(y, \lambda(t))$, so it follows from Lemma A.3 that $b_\eta(\beta)$ is represented by $\tilde{f}$. Note that the image of $\tilde{f}$ lies in $\tilde{W}$ and thus $\Phi_k(\beta)$ is represented by

$$(S^k, s_0) \rightarrow (W', w'_0), \quad y \mapsto \tilde{f}(y).$$

This map factors as the following composition of inclusions

$$S^k \rightarrow D^{k+1} \times \{d_0\} \rightarrow W',$$

which is clearly contractible. Therefore $\Phi_k(\beta) = 0$. Note that all maps in the definition of Φ_k are compatible with the respective π_1 -actions, hence the entire $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by β is contained in the kernel of Φ_k .

Step 2. Next we explain how to kill multiple homotopy classes at once. To this end assume that W is connected. Let $\beta_i \in \pi_k(W, w_i)$ for interior points $w_i \in W$, $1 \leq i \leq l$. Again suppose that the β_i can be represented by embeddings $f_i: (S^k, s_0) \rightarrow (W, w_i)$, which extend to embeddings $F_i: S^k \times D^{n-k} \rightarrow W$ into the interior of W . Moreover assume these embeddings to be disjoint, that is $\text{im}(F_i) \cap \text{im}(F_j) = \emptyset$ for $i \neq j$. This allows us to do surgery along all F_i at once, i.e. we can define

$$\tilde{W} := W - \bigcup_{i=1, \dots, l} \text{int}(\text{im}(F_i))$$

and let W' be defined by the pushout

$$\begin{array}{ccc} \coprod_{i=1, \dots, l} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ \coprod_{i=1, \dots, l} D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

In order to make sense of the statement, that we can kill all homotopy classes β_i at once we would like to consider them as elements in the same group. To achieve this let $d_0 \in S^{n-k-1}$ and $\lambda: [0, 1] \rightarrow D^{n-k}$ be as in Step 1 and set

$$w'_i := F_i(s_0, d_0), \quad \eta_i(t) := F_i(s_0, \lambda(t)).$$

Moreover choose any point $v_0 \in \tilde{W}$ together with paths $\gamma'_i: [0, 1] \rightarrow \tilde{W}$ from $w'_i \rightarrow v_0$. Such paths exist, since W was assumed to be connected and removing (thickened) spheres of codimension at least 2 preserves connectedness. Let

$$\gamma_i := \gamma'_i * \eta_i: [0, 1] \rightarrow W$$

denote the composed path from w_i to v_0 and consider the following commutative diagram.

$$\begin{array}{ccccccc}
\pi_k(W, v_0) & \xrightarrow{\text{id}} & \pi_k(W, v_0) & \xleftarrow{\cong} & \pi_k(\tilde{W}, v_0) & \longrightarrow & \pi_k(W', v_0) \\
b_{\gamma_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong & & b_{\gamma'_i} \uparrow \cong \\
\pi_k(W, w_i) & \xrightarrow[b_{\eta_i}]{\cong} & \pi_k(W, w'_i) & \xleftarrow{\cong} & \pi_k(\tilde{W}, w'_i) & \longrightarrow & \pi_k(W', w'_i)
\end{array}$$

As we have seen in Step 1, the class β_i is contained in the kernel of the bottom horizontal composition, hence $b_{\gamma_i}(\beta_i)$ is contained in the kernel of the top horizontal composition, which defines a map

$$\pi_k(W, v_0) \rightarrow \pi_k(W', v_0).$$

As in Step 1 we conclude that this map is an epimorphism containing the $\mathbb{Z}[\pi_1(W, v_0)]$ -module generated by $b_{\gamma_1}(\beta_1), \dots, b_{\gamma_l}(\beta_l)$ in its kernel. Moreover, the same arguments as in Step 1 apply to show that $\pi_j(W, v_0) \cong \pi_j(W', v_0)$ for $j \leq k-1$.

Step 3. Finally we show how to kill multiple homotopy classes as formulated in (i). Let $g_i: (S^k, s_0) \rightarrow (W, w_0)$ be embeddings with trivial normal bundle representing the class α_i . By a general position argument we can find isotopies $H_i: S^k \times [0, 1] \rightarrow W$ from g_i to embeddings $f_i: S^k \rightarrow W - \{w_0\}$, such that $f_i(S^k) \cap f_j(S^k) = \emptyset$ for $i \neq j$. Moreover we may assume, that the image of f_i is contained in the interior of W . Since the normal bundle of g_i is trivial, the same holds for f_i . Therefore we can extend the f_i to embeddings

$$F_i: S^k \times D^{n-k} \rightarrow W - \{w_0\}$$

into the interior of W , such that $\text{im}(F_i) \cap \text{im}(F_j) = \emptyset$ for $i \neq j$. Now we are in the setting as in Step 2 for $w_i = f_i(s_0)$ and $\beta_i = [f_i] \in \pi_k(W, w_i)$. Define paths $\gamma_i(t) := H_i(s_0, 1-t)$ from w_i to w_0 . Then $b_{\gamma_i}(\beta_i) = \alpha_i$ by Lemma A.3. We would like to apply Step 2 to $v_0 = w_0$, however the paths γ_i are not of the same form as in Step 2 yet. Let us ignore this issue for a moment and see how Step 3 follows. Again let W' be the manifold obtained by surgery along $F_1, \dots, F_l$ as before. Step 2 now yields an epimorphism

$$\Phi: \pi_k(W, w_0) \rightarrow \pi_k(W', w_0).$$

containing the $\mathbb{Z}[\pi_1(W, w_0)]$ -module generated by the $b_{\gamma_i}(\beta_i) = \alpha_i$ in its kernel. Moreover $\pi_j(W, w_0) \cong \pi_j(W', w_0)$ for $j \leq k-1$.

We now construct a homotopy to bring γ_i into the desired form. Let $t_i := \sup\{t \in [0, 1] \mid \gamma_i([0, t]) \subseteq \text{im}(F_i)\}$ be the first time at which γ_i exits $\text{im}(F_i)$. For $t \in [0, t_i]$ we can then write $\gamma_i(t)$ in the form

$$\gamma_i(t) = F_i(\tilde{\sigma}_i(t), \tilde{\lambda}_i(t)).$$

for suitable paths $\tilde{\sigma}_i, \tilde{\lambda}_i$. Now $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ are homotopic to the reparametrized paths

$$\sigma_i(t) = \begin{cases} \tilde{\sigma}_i(0) = s_0, & t \in \left[0, \frac{t_i}{2}\right] \\ \tilde{\sigma}_i(2t - t_i), & t \in \left[\frac{t_i}{2}, t_i\right] \end{cases} \text{ and } \lambda_i(t) = \begin{cases} \tilde{\lambda}_i(2t), & t \in \left[0, \frac{t_i}{2}\right] \\ \tilde{\lambda}_i(t_i), & t \in \left[\frac{t_i}{2}, t_i\right] \end{cases}$$

respectively. So after applying this homotopy we may replace $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ by σ_i and λ_i . For $t \in [0, \frac{t_i}{2}]$, $\gamma_i(t)$ can then be written as

$$\gamma_i(t) = F_i(s_0, \lambda_i(t)),$$

which is of the same form as η_i in Step 2. It remains to bring $\gamma'_i := \gamma_i|_{[\frac{t_i}{2}, 1]}$ into the desired form, meaning the image of γ'_i should be completely contained within $\tilde{W}$. We can achieve this as follows. Applying a general position argument we can assume, that γ'_i does not intersect any of the embedded spheres $f_1(S^k), \dots, f_l(S^k)$. Consider the deformation retraction

$$r: W - \bigcup_{j=1, \dots, l} f_j(S^k) \rightarrow \tilde{W}$$

induced by the deformation retractions $S^k \times (D^{n-k} - 0) \rightarrow S^k \times S^{n-k-1}$. Then $r \circ \gamma'_i$ is a path in $\tilde{W}$, which is homotopic to γ'_i relative to the endpoints. So in conclusion we may indeed assume, that γ_i is of the form as in Step 2.

Step 4. We can now deduce the relative version (ii) of the proposition using (i) applied to the given classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$. Using these classes construct $\tilde{W}$ and W' as before. For $i \leq k$ consider the following diagram, where all maps are induced by inclusions and the rows are exact.

$$\begin{array}{ccccccccc}
\pi_i(M, x_0) & \longrightarrow & \pi_i(W, x_0) & \longrightarrow & \pi_i(W, M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(W, x_0) \\
\uparrow \text{id} & & \uparrow \cong & & \uparrow & & \uparrow \text{id} & & \uparrow \cong \\
\pi_i(M, x_0) & \longrightarrow & \pi_i(\tilde{W}, x_0) & \longrightarrow & \pi_i(\tilde{W}, M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(\tilde{W}, x_0) \\
\downarrow \text{id} & & \downarrow & & \downarrow & & \downarrow \text{id} & & \downarrow \cong \\
\pi_i(M, x_0) & \longrightarrow & \pi_i(W', x_0) & \longrightarrow & \pi_i(W', M, x_0) & \longrightarrow & \pi_{i-1}(M, x_0) & \longrightarrow & \pi_{i-1}(W', x_0)
\end{array}$$

It follows from the Five Lemma that

$$\pi_i(\tilde{W}, M, x_0) \rightarrow \pi_i(W, M, x_0)$$

is an isomorphism and

$$\pi_i(\tilde{W}, M, x_0) \rightarrow \pi_i(W', M, x_0)$$

is an isomorphism for $i \leq k-1$. This shows $\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$ for $i \leq k-1$. If $i = k$ the map $\pi_k(\tilde{W}, x_0) \rightarrow \pi_k(W', x_0)$ is surjective and a diagram chase shows that

$$\pi_k(\tilde{W}, M, x_0) \rightarrow \pi_k(W', M, x_0)$$

is surjective as well. Thus we can define the epimorphism

$$\Phi_{\text{rel}}: \pi_k(W, M, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, M, x_0) \rightarrow \pi_k(W', M, x_0).$$

We have already seen that $\alpha_1, \dots, \alpha_l$ are contained in the kernel of the map

$$\pi_k(W, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, x_0) \rightarrow \pi_k(W', x_0).$$

So commutativity of the above diagram implies

$$\Phi_{\text{rel}}(\alpha_i^{\text{rel}}) = 0.$$

Again all maps in the definition of Φ_{rel} are compatible with the respective π_1 -actions, such that Φ_{rel} contains the entire $\mathbb{Z}[\pi_1(M, x_0)]$ -module generated by $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ in its kernel. $\square$

- gucken ob konsistent mit sowas wie manifold oder smooth manifold, psc metric, psotiive scalar curvature metric etc.
- formatierungs shit (schauen, dass keine weirden zeilenumbrüche usw.)